



BIRZEIT UNIVERSITY

Faculty of Engineering and Technology

Department of Computer Science

COMP333—DATA BASE SYSTEMS

Database Project

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1 Group Name

Bawabet Al-Quds Roastery

2 Group Members

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3 Project Title

Bawabet Al-Quds Roastery: An Integrated Sales, Purchase, and Warehouse Management System

4 Project Description

The primary goal of this project is to develop a robust, database-driven application for **Bawabet Al-Quds Roastery** using **Java** for the application layer and **MySQL** as the relational database management system. The system is designed to digitize and streamline the roastery's high-volume operations by integrating three core pillars of their business:

- **Sales System:** A comprehensive Point of Sale (POS) module that manages various transaction types. It will support multi-tier pricing logic based on the customer category (Retail vs. Wholesale) and handle weight-based product calculations with high precision.
- **Purchase Management:** A module dedicated to the procurement of raw materials and supplies. It will track financial and physical transactions with suppliers, ensuring that the cost of goods and supply history are maintained for accurate financial reporting.
- **Warehouse and Inventory Control:** A centralized system to manage the roastery's physical stock. The database will synchronize with the sales and purchase modules to provide real-time updates on stock availability across different categories, preventing shortages and optimizing storage.

By replacing manual bookkeeping with this centralized MySQL solution, the project ensures data integrity, facilitates advanced data querying for business insights, and provides a professional GUI that enhances the user experience for the roastery staff.

5 Client Information

Client Name: Bawabet Al-Quds (Jerusalem Gate) Roastery

Business Type: Retail and Wholesale Distribution of Specialty Coffee and Roasted Goods

Location: Ramallah Palestine

5.1 Organizational Scope and Transactions

Bawabet Al-Quds is an organization characterized by substantial physical and financial transactions, serving a wide variety of clients. The system is built to accommodate the specific needs of three primary client segments:

- **Retail Individuals:** Walk-in customers purchasing smaller, customized quantities of roasted products.
- **Wholesale Salesmen:** Independent distributors who buy bulk quantities for re-distribution, requiring specialized bulk pricing and transaction history tracking.
- **Supermarkets and Grocers:** Large Business-to-Business clients who receive regular shipments and require managed accounts for long-term invoicing and credit tracking.